

# Cambridge International AS & A Level

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## MATHEMATICS

**9709/33**

Paper 3 Pure Mathematics 3

October/November 2023

**1 hour 50 minutes**

You must answer on the question paper.

You will need: List of formulae (MF19)

## INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

## INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.

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- 1** Find the set of values of  $x$  satisfying the inequality  $|2^{x+1} - 2| < 0.5$ , giving your answer to 3 significant figures. [4]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

- 2 On an Argand diagram, shade the region whose points represent complex numbers  $z$  satisfying the inequalities  $|z - 1 + 2i| \leq |z|$  and  $|z - 2| \leq 1$ . [5]

- 3** The polynomial  $2x^3 + ax^2 + bx + 6$ , where  $a$  and  $b$  are constants, is denoted by  $p(x)$ . When  $p(x)$  is divided by  $(x + 2)$  the remainder is  $-38$  and when  $p(x)$  is divided by  $(2x - 1)$  the remainder is  $\frac{19}{2}$ .

Find the values of  $a$  and  $b$ .

[5]

[illegible]

- 4 Solve the quadratic equation  $(3 + i)w^2 - 2w + 3 - i = 0$ , giving your answers in the form  $x + iy$ , where  $x$  and  $y$  are real. [5]

[illegible]

- 5** Find the exact coordinates of the stationary points of the curve  $y = \frac{e^{3x^2-1}}{1-x^2}$ . [6]

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- 6** (a) Show that the equation  $\cot^2 \theta + 2 \cos 2\theta = 4$  can be written in the form

$$4 \sin^4 \theta + 3 \sin^2 \theta - 1 = 0. \quad [3]$$

[illegible]

**[Turn over**

**7** The equation of a curve is  $x^3 + y^2 + 3x^2 + 3y = 4$ .

(a) Show that  $\frac{dy}{dx} = -\frac{3x^2 + 6x}{2y + 3}$ . [3]

[illegible]

- (b)** Hence find the coordinates of the points on the curve at which the tangent is parallel to the  $x$ -axis. [5]

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- 8 The variables  $x$  and  $y$  satisfy the differential equation

$$e^{4x} \frac{dy}{dx} = \cos^2 3y.$$

It is given that  $y = 0$  when  $x = 2$ .

Solve the differential equation, obtaining an expression for  $y$  in terms of  $x$ .

[7]

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**9** Let  $f(x) = \frac{17x^2 - 7x + 16}{(2 + 3x^2)(2 - x)}$ .

(a) Express  $f(x)$  in partial fractions.

[5]

[illegible]

- (b) Hence obtain the expansion of  $f(x)$  in ascending powers of  $x$ , up to and including the term in  $x^3$ . [5]

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- (c) State the set of values of  $x$  for which the expansion in (b) is valid. Give your answer in an exact form. [1]

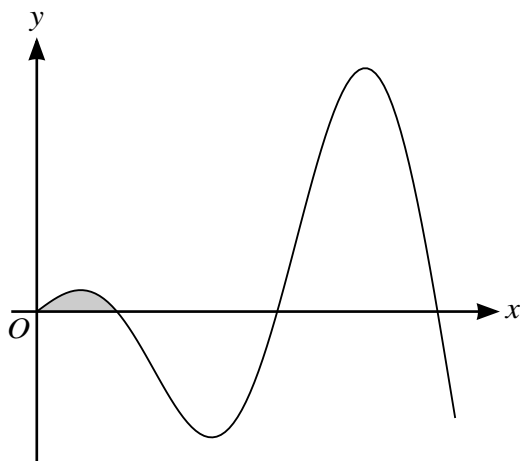
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**(a)** Find the equation of the tangent to the curve at the point where  $x = \frac{1}{2}\pi$ . [4]

[illegible]

- (b) Find the exact area of the shaded region shown in the diagram, bounded by the curve and the  $x$ -axis. [5]

[illegible]

- 11** The line  $l$  has equation  $\mathbf{r} = \mathbf{i} - 2\mathbf{j} - 3\mathbf{k} + \lambda(-\mathbf{i} + \mathbf{j} + 2\mathbf{k})$ . The points  $A$  and  $B$  have position vectors  $-2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$  and  $3\mathbf{i} - \mathbf{j} + \mathbf{k}$  respectively.

(a) Find a unit vector in the direction of  $l$ . [2]

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The line  $m$  passes through the points  $A$  and  $B$ .

(b) Find a vector equation for  $m$ . [2]

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[illegible]

[illegible]

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